

ARTICLE

Reproductive effort and terminal investment in a multispecies assemblage of Amazon electric fish

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Abstract

The terminal investment hypothesis (TIH) predicts that individuals with favorable prospects for future reproduction (i.e., high residual reproductive value, RRV) should moderate current reproductive investment in favor of growth, survival, and future reproduction, whereas those with low RRV should “terminally invest” by diverting somatic resources towards current reproduction at the expense of future reproduction. However, support for the TIH in wild animal populations is fragmentary, and the ecological contexts of terminal investment remain poorly known. We report a remarkable case of simultaneous terminal investment involving five sympatric species of the electric knifefish genus *Brachyhypopomus*, from Amazonian floodplain and terra firme stream habitats. We found that terminal investment is synchronized by seasonal breeding, in response to circannual environmental variation in mortality risk. Four species exhibit a uniseasonal iteroparous (annual) life history with complete post-reproductive mortality after a single breeding season. One species (*Brachyhypopomus beebei*) exhibits a 2-year multiseasonal iteroparous life history with breeding in two seasons and post-reproductive mortality after the second. In mature females and (most) males of the annual species, as well as in both mature female and male *second-year* (but not *first-year*) *B. beebei*, we documented an increase in two metrics of reproductive effort (size-adjusted gonad mass and electric signal amplitude) and a concomitant reduction in somatic condition (size-adjusted somatic mass), all in response to proximity to the end of the common breeding season, when RRV approximates zero. In mature *first-year B. beebei*, we documented neither an increase in reproductive effort nor a decline in somatic condition, implying an alternative strategy of reproductive restraint. Our findings support Kirkwood’s disposable soma theory, which posits that death by reproductive exhaustion can be delayed if terminal investment is replaced by reproductive restraint, allowing individuals to survive and breed in a subsequent season. Deferral of the terminal investment response in annual species, and the origin of a gonadal regression-regeneration sequence, may open pathways for rapid evolutionary transitions to multiseasonal iteroparity. Excepting the age (year-group) dependency of terminal investment in *B. beebei*, we were unable to identify

intrinsic cues or extrinsic environmental cues for the terminal investment response in *Brachyhypopomus*.

KEYWORDS

Brachyhypopomus, disposable soma theory, electric organ discharge, Gymnotiformes, iteroparity, life history ecology, reproductive ecology, reproductive investment, residual reproductive value, seasonality, semelparity, sensory ecology

INTRODUCTION

Animals that breed repeatedly must reach an optimal trade-off between the allocation of limited resources to current reproduction, which comes at the expense of survival and future reproductive success, versus the allocation of resources to somatic growth and maintenance, which comes at the expense of current reproduction, but improves future reproductive success (Roff, 1992; Stearns, 1992). Life history theory predicts that iteratively breeding animals can maximize lifetime reproductive success by modulating reproductive allocation as a function of their residual reproductive value (RRV); that is, their prospects for future reproduction (Pianka & Parker, 1975; Roff, 1992; Stearns, 1992; Williams, 1966). The “terminal investment hypothesis” (TIH; Clutton-Brock, 1984) formalizes this prediction by postulating that when the RRV of reproductive individuals is high, investment in costly current reproduction should be moderated in favor of growth, maintenance, and survival, thereby increasing the potential for future reproductive events. Conversely, when RRV is low, the TIH predicts an unsustainable elevation of reproductive effort, in which resources are allocated to current reproduction at rates exceeding previous reproductive events, and at the expense of future survival and reproduction (Charlesworth & León, 1976; Clutton-Brock, 1984; Schaffer, 1974; Williams, 1966). A reproductive adult’s RRV typically depends on age, condition, and immune status, as well as extrinsic ecological factors predictive of mortality risk such as food and substrate availability, predator densities, and pathogen infection rates (Duffield et al., 2017; Roff, 2002; Stearns, 1992). We use the term “terminal investment” to refer to the process of intensified reproductive allocation in response to a lowering of RRV; see for example Pärt et al. (1992) and Duffield et al. (2017) for a similar approach.

Some of the most persuasive evidence for the TIH derives from experiments that manipulate an animal’s RRV, for example by simulated or actual pathogenic infection (e.g., Minchella & Loverde, 1981; Velando et al., 2006), by food deprivation (e.g., Blanckenhorn et al., 2008; Gavassa & Stoddard, 2012), or by altering the perception of predation risk (e.g., Candolin, 1998). In these experiments, older individuals tend to respond to a reduction in RRV

by diverting somatic resources to current reproduction. Instead, younger individuals typically moderate current reproduction in favor of growth, maintenance, and survival. Nonetheless, support for the TIH from non-manipulative field studies of wild animal populations is fragmentary, and the ecological contexts of terminal investment consequently remain incompletely understood (Duffield et al., 2017; Javoiš, 2013). In fact, many field-based studies report no age-specific modulation of reproductive effort at all (Duffield et al., 2017), or a pattern of *reduced* reproductive effort in older individuals consistent with senescence (Broussard et al., 2003; Loison et al., 1999; McNamara et al., 2009). In many other field-based studies, support for the TIH is equivocal (Duffield et al., 2017; Javoiš, 2013; Velando et al., 2006). Those studies that do provide convincing evidence for the TIH in wild populations, reviewed by Duffield et al. (2017), have focused mostly on the effects of age, or the interaction of age with pathogenic infection.

The role of natural abiotic and biotic environmental variation as a potential regulator of terminal investment has, in contrast, received relatively little attention, despite the well-known effects of seasonal variation (e.g., in resource availability and predation) on the timing of reproductive effort (Baker, 1938). Many short-lived iteroparous animals, including some groups of insects, fish, and other taxa with annual life histories occur in highly seasonal environments. These taxa might therefore be expected to modulate terminal investment in response to changing seasonal environmental conditions or other cues predictive of the end of their reproductive lives.

To explore the relationship between terminal investment and seasonal environmental variation, we describe circannual patterns of reproductive investment in a sympatric multispecies assemblage of the electric knifefish genus *Brachyhypopomus*, from the upper Amazon. *Brachyhypopomus* is an attractive model system for exploring the effects of environmental variation on reproductive life-history trade-offs because, as with other fish, traits informative of the allocation of energy to reproduction versus somatic growth and maintenance are readily quantified from preserved specimens via size-adjusted metrics of gonad mass and somatic mass. Additionally,

Brachyhypopomus generate a constant, low-voltage, pulsed electric organ discharge (EOD) for electrolocating objects in the dark and for reproductive signaling (in both mature males and females). The amplitude of the EOD signal is positively correlated to the signal's metabolic energy cost (Salazar & Stoddard, 2008) and therefore serves as a quantifiable metric of energetic investment in reproductive signaling.

We previously reported that *Brachyhypopomus* inhabit upper Amazonian floodplain lakes and terra firme rainforest streams where they breed iteratively (via fractional spawning events in females) during an annual wet-season breeding period (Waddell et al., 2019; Waddell & Crampton, 2020). Breeding begins during a period of rapid expansion of new habitat associated with the rise of river level (for floodplains) or local rainfall (for streams) and ends with the onset of a dry-season period of elevated predation and reduced food availability, which is unfavorable for larval recruitment and adult survival (Crampton, 2011; Macedo et al., 2015; Waddell & Crampton, 2020). Using circannual size-frequency distributions and analyses of sagittal otolith growth annuli, we also established that four common *Brachyhypopomus* species (*B. bennetti*, *B. sullivani*, *B. verdii*, *B. walteri*) have a 1-year uniseasonal iteroparous life history (sensu Kirkendall & Stenseth, 1985) (i.e., “annual” life history), with a protracted single breeding season followed by complete post-reproductive mortality (Waddell et al., 2019). A fifth species, *Brachyhypopomus beebei*, has a 2-year multi-seasonal iteroparous (i.e., “biseasonal”) life history, with repeated breeding in both a protracted first and second reproductive season, and complete post-reproductive mortality following the second season (Waddell et al., 2019).

Life-history theory predicts that uniseasonal *Brachyhypopomus* species should terminally invest in response to proximity to the end of their single breeding season (at the onset of the annual dry season) because this represents the moment at which their RRV declines to zero (Pianka & Parker, 1975; Poizat et al., 1999). Therefore, in annual species, we would predict an intensified diversion of resources from the soma to reproduction in late-season reproductive events relative to early-season reproductive events. In contrast, species with a biseasonal life history are predicted to show reproductive restraint in their first breeding season but terminally invest in their second (final) season (Billman & Belk, 2014; Fisher, 1930; Pianka & Parker, 1975; Williams, 1966). More specifically, biannual species are predicted to terminally invest towards the end of their second season, via the same intensification of late-season reproductive effort (relative to early season effort) predicted for uniseasonal species. As conceptualized in Figure 1, we test these predictions using size-adjusted measures of reproductive effort and somatic mass from individuals that were captured and preserved

throughout the annual hydrological cycle and using a simple estimate of an individual fish's RRV: time from capture until the end of its final breeding season. Our tests of the TIH in upper Amazon *Brachyhypopomus* are based on population-level responses and assume that the sum of individual responses across a population is representative of temporal changes in resource apportioning in individual fish (Billman & Belk, 2014; Poizat et al., 1999).

Using this approach, we explore whether seasonal environmental variation and associated circannual breeding periodicity can produce a synchronized multispecies pattern of terminal investment in which metrics of reproductive investment increase as a function of declining RRV, while somatic condition concomitantly decreases, indicating a future “survival cost” (sensu Fisher & Blomberg, 2011; i.e., a reduction in the probability of future survival and reproduction). The occurrence of multiple *Brachyhypopomus* species in two types of highly seasonal habitats, the presence of species with both uniseasonal and biseasonal iteroparous life histories, and the occurrence of costly reproductive signals provide a unique set of opportunities to explore the terminal investment response in natural animal populations.

MATERIALS AND METHODS

Study taxon

Brachyhypopomus (Hypopomidae, Gymnotiformes) are small (mostly <300 mm total length) nocturnally active predators of invertebrates (Albert & Crampton, 2003; Crampton et al., 2016a). Females are oviparous, with iterative fractional spawning and brood hiding (sensu Balon, 1975), and neither sex undertakes parental care (Waddell et al., 2019). All species in our study area were included in a recent species-level taxonomic revision of the genus (Crampton et al., 2016a) and a recent phylogenetic reconstruction (Crampton et al., 2016b).

Study sites

Sampling was conducted near Jenaro Herrera, Peru, in 2013–2014 (Figure 2). We sampled floating macrophyte meadows in two floodplain lakes sites (F1–F2) of the Ucayali River Holocene inundation plain, both of which had year-round water content during the high-amplitude (~10 m) inundation cycle (Figure 2g); other lakes in the area could not be sampled year-round because they dried out at low water. The floating macrophyte meadows expanded to reach their maximum extent at high water (Figure 2d) and contracted into small refuges at low water but were occupied all year by *Brachyhypopomus*

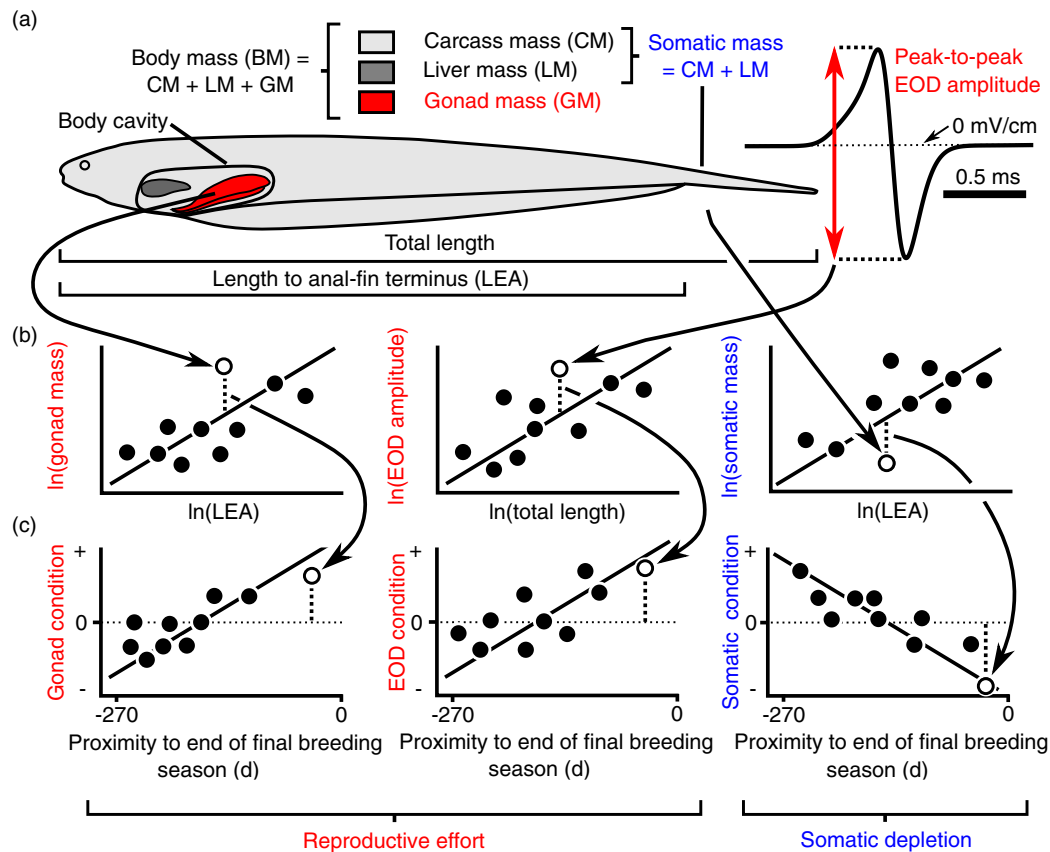


FIGURE 1 A test for terminal investment. (a) Body mass, length, and electric organ discharge (EOD) amplitude measurements in the electric fish *Brachyhyppomus*. (b) Linear regressions to calculate gonad condition (size-adjusted gonad mass), EOD condition (size-adjusted EOD amplitude), and somatic condition (size-adjusted somatic mass); white dot represents the single measured fish in panel a; black dots represent all other individuals of the same species, sex, and year group. (c) Under conditions of terminal investment, gonad condition and EOD condition, which are metrics of reproductive condition, are expected to increase as a function of proximity to the end of a species' final breeding season, that is, time from capture of a specimen to the end of its final breeding season. In contrast, somatic condition, a metric of somatic depletion, is expected to decrease. Example here is for a hypothetical 9-month (270-d) breeding season

(Waddell & Crampton, 2020). We also sampled four small (<4 m wide) rainforest stream sites (T1–T4) in Paleogene–Neogene terra firme formations above the limit of seasonal river-floodplain inundation (Figure 2g). In response to a low-amplitude (<1 m) seasonal flood regime, these streams inundated low-lying adjacent terrain to form shallow pools during the rainy season (Figure 2f) but were confined to their main courses during the dry season (Figure 2e; Waddell & Crampton, 2020). All sites were located in intact primary forest within protected areas: Reserva Nacional Pacaya Samiria (sites F1–F2), Jenaro Herrera Ecological Station, Instituto de Investigación de la Amazonía Peruana (sites T1–T3); and private land (site T4).

Sampling

Each site was sampled 1–10 times (usually 2–6 times) per month, for 12 months, with a total effort across all six

sites of 193 sampling events and 937 h (Waddell et al., 2019). All sampling was conducted at night, from the end of astronomical twilight until as late as 02:00, during the period of peak electric fish activity. Abundances were quantified by timed capture-per-unit effort (CPUE), as individuals/h, using a custom-made electric fish finder to locate *Brachyhyppomus* (Haag et al., 2019) and a dipnet to then capture them (Regular Hex Trapnet with 3 mm mesh and 30.5 cm bag depth; Duraframe Dipnet, Viola, Wisconsin, USA). The use of electric fish finders allowed close to 100% capture efficiency for all individuals (excepting larval and small post-larval stages); see Waddell et al. (2019) for further sampling details. Following EOD amplitude measurements (see *Electric signal amplitude*), all fish were kept without feeding for 48 h in individual buckets or air permeable polymer bags (Kordon LLC, San Francisco California, USA), to fully void their alimentary canals. They were then euthanized in 600 mg/L eugenol and fixed in 3.7% formalin. All

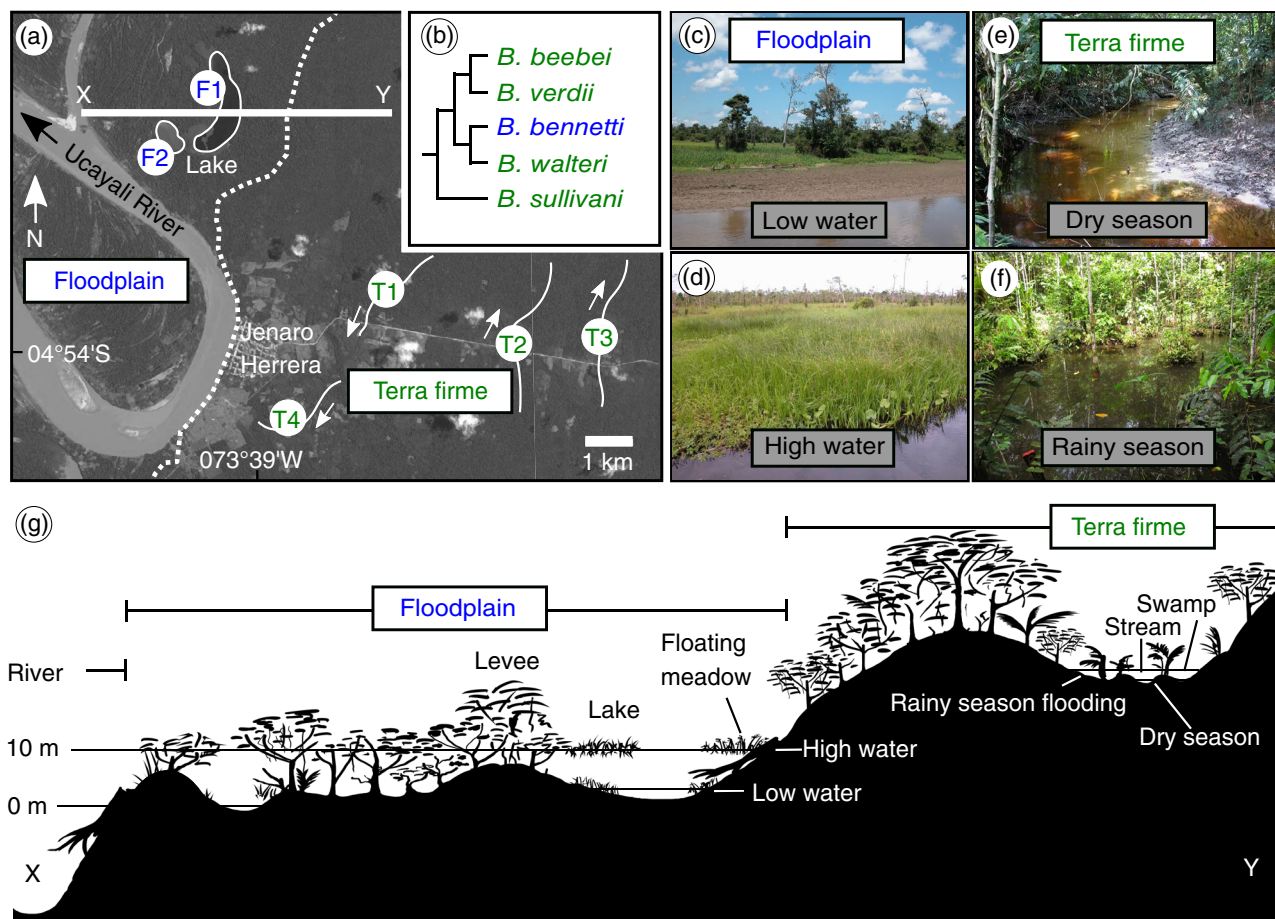


FIGURE 2 Upper Amazon study area, habitats, and *Brachyhypopomus* assemblage. (a) Map of area; dotted white line is river-floodplain-terra firme boundary. Solid white lines represent sample sites. (b) Phylogeny and habitat occupancy of *Brachyhypopomus* following Crampton et al. (2016b): blue, floodplain; green, terra firme stream. (c) Floodplain lake at low water. (d) Floodplain lake with floating macrophyte meadows at high water. (e) Terra firme stream confined to main course at low water. (f) Ephemeral swamp adjacent to terra firme stream in rainy season. (g) Schematic cross-section from X–Y transect in panel a; 0 and 10 m river levels for river floodplain represent flood minimum and maximum from Waddell and Crampton (2020)

measurements of length and body and organ mass were taken from specimens that were formalin fixed for 7–10 d, before subsequent preservation in 70% ethanol. We excluded from our analyses individuals in which $\geq 10\%$ of estimated intact total length was missing due to caudal filament loss, following Waddell et al. (2019). Specimens were identified using the diagnoses and descriptions in Crampton et al. (2016b). Animal use was approved by University of Central Florida Institutional Animal Care and Use Committee (protocols 11–39, 14–22).

Maturation staging

We included in our analyses only spawning capable (gonad stage 3 and 4) individuals, following a modified version of the Brown-Peterson et al. (2011) maturational staging scheme (Waddell et al., 2019; Waddell & Crampton, 2018).

In so doing, we included all reproductive individuals that had begun to breed but excluded fish with spent gonads; these were mostly restricted to resting individuals of the multiseasonal iteroparous species *B. beebei* (Waddell et al., 2019). For fish with fractional spawning, stage 4 (actively spawning) females are those in the process of releasing a ripe batch of oocytes, which occurs over several days in *Brachyhypopomus* (Kirschbaum & Schugardt, 2002), while stage 3 (non-actively spawning) females are those in the period between successive spawning events, or those that are about to spawn for the first time (Nuñez & Duponchelle, 2009).

Reproductive periods

We defined the breeding period of each species as when we sampled at least one spawning capable individual

(of either sex), from at least one sampling site, during uninterrupted consecutive 1-week periods.

Electric signal amplitude

All EOD measurements were made within 10 minutes (usually <5 minutes) of a fish's capture using a recording apparatus that we moved along the stream/lake margin. Each fish was held temporarily in a nylon mesh envelope suspended mid-tank, in a modified 114-L cooler (Igloo Polar 120, Igloo Products Corp., Katy, Texas), with its long axis parallel to custom-built tank-end Ni-Cr electrodes, following Waddell et al. (2016). The cooler was filled with water from the capture site adjusted to a standard conductivity of $55 \pm 2 \mu\text{S/cm}$ and temperature to $27 \pm 0.1^\circ\text{C}$, thereby eliminating EOD amplitude variation resulting from inter- and intrasite or habitat variation in conductivity and temperature (e.g., variation along the annual hydrological cycle). We obtained a standardized measure of the peak-to-peak amplitude of the head-to-tail recorded EOD, which we report as the equivalent field intensity in mV/cm (to $\pm 0.2 \text{ mV/cm}$) at 10 cm from the fish's center (midpoint between snout and caudal filament terminus), following Franchina and Stoddard (1998). By obtaining EOD recordings only after astronomical twilight, we allowed time for the EOD to fully transition from day to night amplitudes, via natural hormonally mediated circadian rhythms (Stoddard et al., 2007). By performing all recordings within 10 minutes of capture, we ensured that EOD amplitude was undisrupted by the action of stress hormones released during capture and handling; these hormones affect the EOD, via changes in electrocyte membrane physiology, no earlier than 10 minutes after the onset of a stress stimulus (Allee et al., 2008).

Measurements from preserved specimens

As summarized in Figure 1, we measured total length (TL) and length from snout to anal-fin terminus (LEA) to $\pm 1 \text{ mm}$ using a ruler, and total body mass (BM), gonad mass (GM), and liver mass (LM), to $\pm 1 \text{ mg}$, using a Sartorius 64-1S balance (Sartorius AG, Göttingen, Germany). We then calculated "somatic mass" (SM) as $\text{BM} - \text{GM}$. To document sex variation in fish liver function (Allen & Wootton, 1982; Bruslé & González i Anadon, 1996) and to evaluate the role of energy stores in the liver vs. in the skin, viscera, or other somatic locations, we also calculated a metric of "carcass mass" (CM) as $\text{BM} - (\text{GM} + \text{LM})$. Variation in SM and CM due to food ingestion was eliminated by ensuring the alimentary tracts of all specimens were completely empty (see *Sampling*). We quantified relative overall body condition

for each individual by calculating the scaled mass index (SMI), following Peig and Green (2009). The SMI is a modification of conventional residual condition indices that is based on residuals from standard major axis (SMA) $\log(\text{body mass}) \times \log(\text{body length})$ regressions, corrected to a standardized body length.

Diagnosing terminal investment

Based on data collected for all individuals of a given combination of species, sex, and year group, we diagnosed terminal investment by the simultaneous occurrence of two patterns, as summarized in Figure 1: First, an increase in one or both of two metrics of "reproductive effort" as a function of proximity to the end of the breeding season: (1) size-adjusted gonad mass (gonad condition, GC); (2) size-adjusted EOD amplitude (EOD condition, EodC). Second, a simultaneous decrease in size-adjusted somatic mass (somatic condition, SC) as a function of proximity to the end of the breeding season. We quantified proximity to the end of the breeding season as the number of days between an individual's capture, and the end of its breeding season, as defined in *Reproductive periods* and expressed as a negative value. The first- (0+) and second-year (1+) groups of *B. beebei* were divided at 115 mm LEA, following Waddell et al. (2019). To better understand the pathways by which somatic resources are diverted toward reproduction, we also investigated the relationship between proximity to the end of the breeding season and size-adjusted LM (liver condition, LC) and size-adjusted CM (carcass condition, CC; recalling that CM excludes gonad and liver mass). We henceforth refer to GC, EodC, SC, LC, and CC as "condition factors."

The five condition factors were calculated for each individual as follows: (1) GC as the residuals from $\ln(\text{GM}) \times \ln(\text{LEA})$ SMA regression, with $\ln$ -transformation of both axes to linearize the exponential growth relationship (Figure 1); we used SMA regression here to account for allometric scaling, following the rationale of Warton et al. (2006). (2) SC as the residuals from $\ln(\text{SM}) \times \ln(\text{LEA})$ SMA regression (Figure 1). (3) LC as the residuals from $\ln(\text{LM}) \times \ln(\text{LEA})$ SMA regression. (4) CC as the residuals from $\ln(\text{CM}) \times \ln(\text{LEA})$ SMA regression. We used LEA (rather than TL) as the metric of body length in calculating GC, SC, CC, and LC, because of its larger coefficient of correlation with body mass than TL; the weaker correlation for TL arises from intraspecific variation in caudal filament length. (5) EodC as the residuals from a $\ln(\text{EOD amplitude}) \times \ln(\text{TL})$ ordinary least squares (OLS) regression, with $\ln$ transformation of both axes to linearize the known exponential relationship between EOD amplitude and TL (Gavassa et al., 2013) (Figure 1); we used TL

instead of LEA to calculate EodC because the total body length of electric fish approximates the dipole separation of the EOD field, which by Coulomb's law is linearly proportional to the electric field strength (Stoddard et al., 1999).

Finally, we adopted two approaches to detect a terminal investment response. First, as summarized in Figure 1c, for each combination of species, sex, and year group (year group only for *B. beebei*) we generated OLS regressions for each condition factor as a function of proximity to the end of a species' final breeding season. Second, we constructed linear mixed models (LMMs) to explore whether the responses of a given condition factor to proximity to the end of the breeding season were generalized across species, or across both species and sex (with species, sex, or sex-nested-in-species, assigned as random effects).

Linear mixed models

We used LMMs to explore the relationship between each condition factor and proximity to the end of the breeding season, using lme4 (Bates et al., 2021). We estimated F values and associated degrees of freedom by the Satterthwaite approximation method, using lmerTest (Kuznetsova et al., 2017). Marginal R^2 values were calculated following Nakagawa and Schielzeth (2013) using MuMIn (Bartón, 2018) and conditional R^2 values were calculated following the method of Xu (2003). Because our analyses involved only five species, we did not consider phylogenetic LMMs. We removed the 0+ year group for *B. beebei* from all models because we had no a priori expectation of a terminal investment-like response in this year group (this expectation was confirmed by our data, see *OLS regressions: terminal investment*). Following Zuur et al. (2013), we used model selection, with Akaike's information criterion, to determine that the addition of habitat type + site replicate (or site nested in habitat) had negligible effect on model performance. Consequently, we excluded habitat type (floodplain lake/stream) and site replicate (i.e., F1–F2, T1–T4) as variables in all models.

RESULTS

Quantitative sampling results and circannual breeding periodicity

We collected 3,209 *Brachyhypopomus* from five common species, of which 1,135 were spawning capable; see Figure 3 for sample sizes and circannual CPUE abundance. One species was found only in floodplain lakes

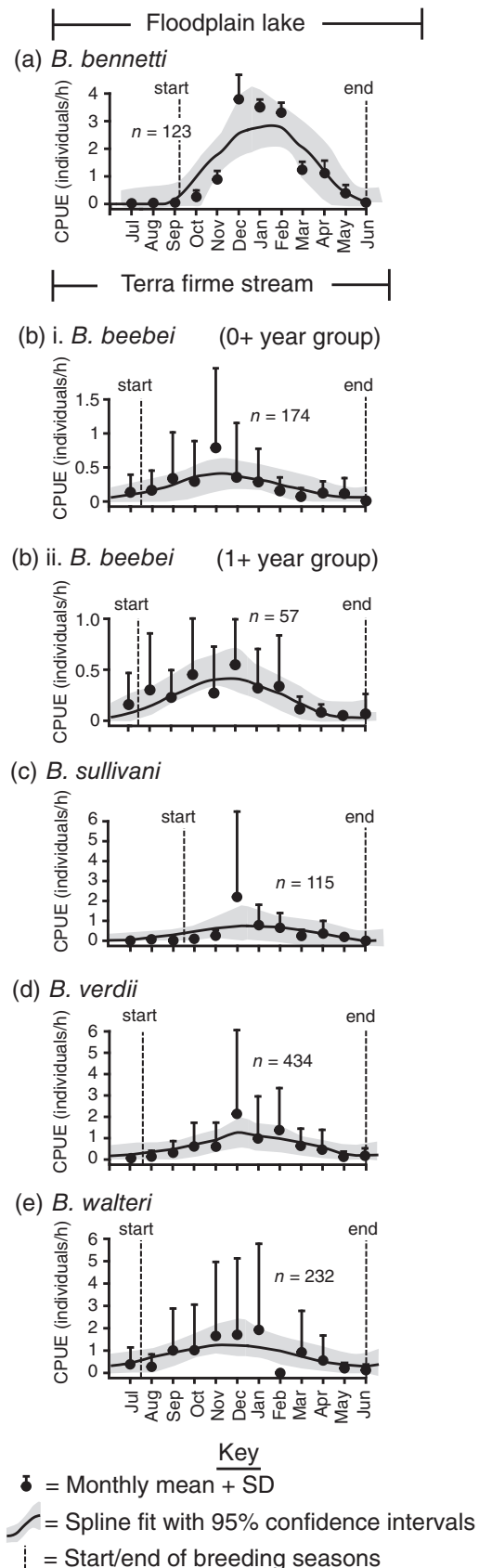


FIGURE 3 Seasonal variation in monthly capture per unit effort abundance of spawning-capable females and males of five *Brachyhypopomus* species

(*B. bennetti*) and three only in terra firme streams (*B. sullivani*, *B. verdii*, *B. walteri*). A fifth species, *B. beebei*, was abundant in streams ($n = 548$) but also found occasionally in floodplain lakes ($n = 11$); we excluded the floodplain specimens from our analyses. Breeding began between mid-July and mid-September and ended coincidentally, in all species, on the following 1 June, ± 1 week (Figure 3). Breeding season lengths were 8.5 months in *B. sullivani*, 8.75 months in *B. bennetti*, and 10.5 months in *B. beebei*, *B. verdii*, and *B. walteri*. The ends of the breeding periods also corresponded to the approximate minima of splines fitted to the monthly CPUE abundance of spawning-capable individuals (Figure 3).

EOD amplitude

Conductivity-standardized peak-to-peak EOD amplitude increased as an exponential function of TL in spawning capable females and males of all five species, including those with EOD waveforms of monophasic structure (*B. bennetti*); biphasic structure (*B. beebei*, *B. verdii*, *B. walteri*); and tetraphasic structure (*B. sullivani*) (Figure 4).

OLS regressions: terminal investment

Both sexes of all four unisexual iteroparous species of *Brachyhypopomus* (with the exception of male *B. walteri*) exhibited a pattern consistent with the expectations of the TIH, in which at least one measure of reproductive effort (i.e., gonad condition and/or EOD condition) increased as a function of proximity to the end of the shared breeding season, while somatic condition simultaneously decreased (Figure 5). Also consistent with theoretical predictions, in the multiseasonal iteroparous species *B. beebei*, we observed a clear pattern of increasing reproductive effort and decreasing somatic condition as a function of proximity to the end of the final breeding season in both sexes of the 1+ group. In contrast, in both sexes of the 0+ year group, we observed no temporal modulations of resource allocation through the breeding season.

Several of the regressions in Figure 5 had strong slope gradients and relatively large coefficients of correlation (R^2), notably in *B. bennetti*, and in the 1+ year group of *B. beebei*. Nonetheless, in some other regressions, even where P values were small, the R^2 values were low and/or the slope gradients were shallow. Despite these variable effect sizes, what is striking about our results is that the slopes were consistently in the direction expected by terminal investment theory. Of the 30 regressions that pertain to a species' final breeding season (i.e., all except those for the 0+ year group of *B. beebei*), 24 (80%) had

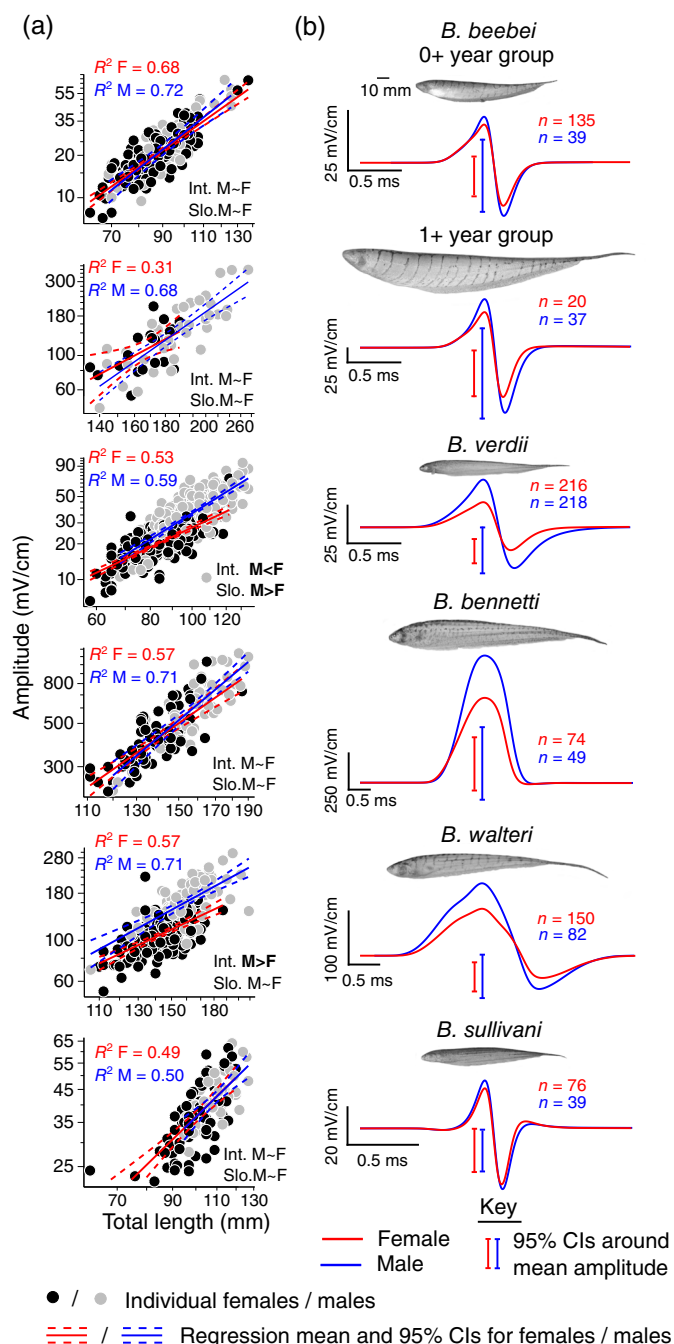


FIGURE 4 Species and sex variation in electric signal (electric organ discharge, EOD) amplitude and waveform for spawning-capable individuals of five *Brachyhypopomus* species. (a) Ordinary least-squares regressions of $\ln(\text{conductivity-standardized peak-to-peak EOD amplitude})$ (as equivalent field intensity in mV/cm at 10 cm from fish's center) as a function of $\ln(\text{total length})$; axis labels represent back-transformed values. Disparities between females (F) and males (M) for intercept (Int.) and slope (Slo.) are by general linear model (boldface type denotes $P < 0.001$). (b) Time-voltage waveforms plotted head-positive upward and scaled to mean peak-to-peak amplitude (for males and females, separately). For plots of all original individual recordings see Appendix S2: Figure S1. Photographs of fish are scaled to maximum body size for each species/year-group

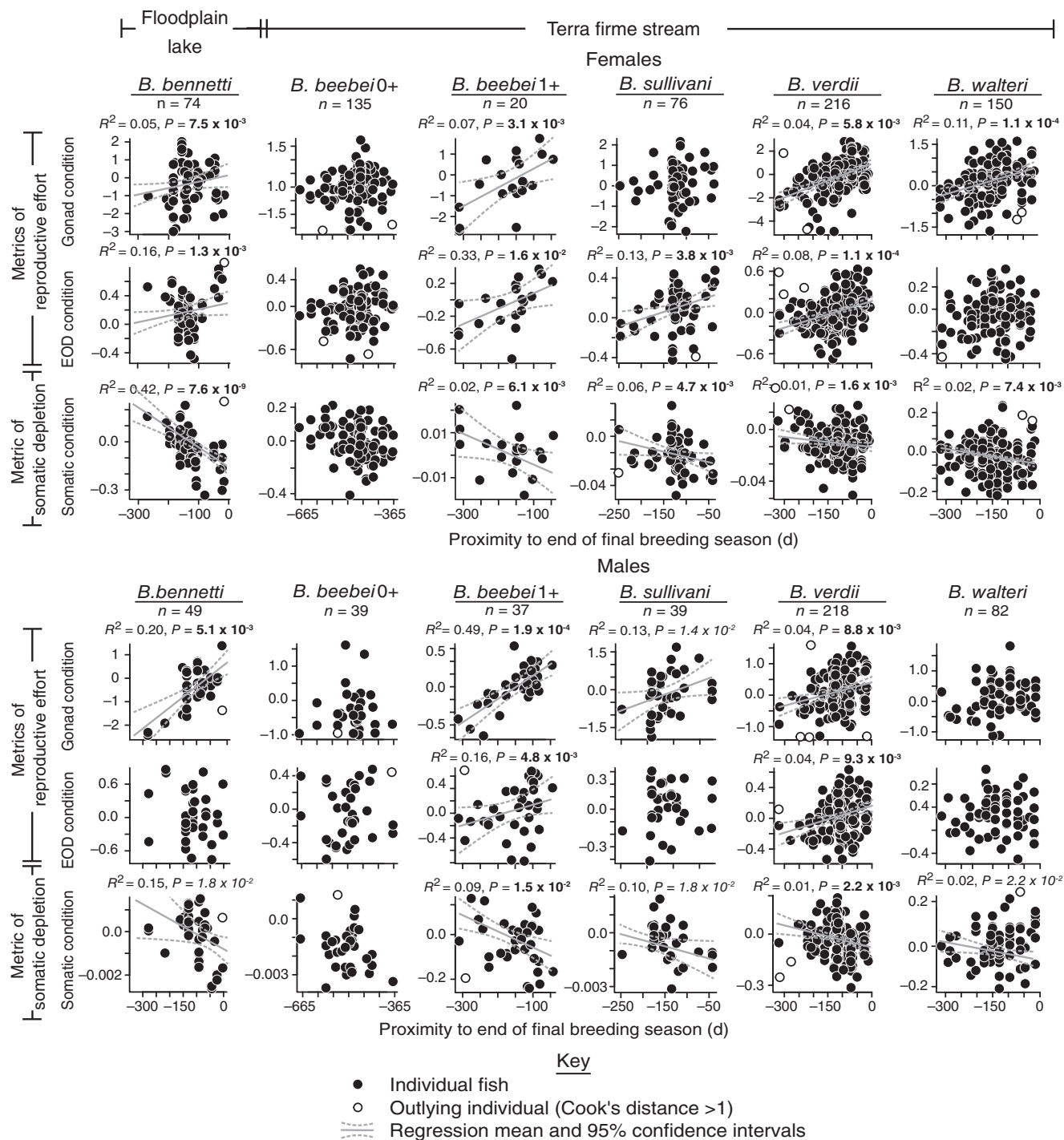


FIGURE 5 Ordinary least-squares regressions of gonad condition, EOD condition, and somatic condition as a function of proximity to end of the final breeding period for spawning-capable females and males of five *Brachyhypopomus* species. Species/year groups with underlined titles exhibit a pattern of resource reallocation consistent with terminal investment, manifest as a population-wide increase in at least one measure of reproductive effort (gonad condition, EOD condition) and a concomitant decrease in somatic condition in response to a critical diminution of residual reproductive value (RRV). Outliers are included in the regression statistics; some outliers are outside axis ranges. Regression lines are reported only for regressions with $P < 0.05$; with boldface indicating $P < 0.05$ and italic indicating $P > 0.05 < 0.1$ following Bonferroni correction ($k = 5$, for the five condition factors of a given sex: gonad condition, EOD condition, somatic condition, liver condition, carcass condition)

significant slopes in the expected direction (positive or negative), but none had slopes in the unexpected direction (Figure 5). More specifically, for somatic condition

the expected *negative* slope was observed in 10 out of 10 regressions (both sexes of all five species). For gonad condition and EOD condition, the expected *positive* slope

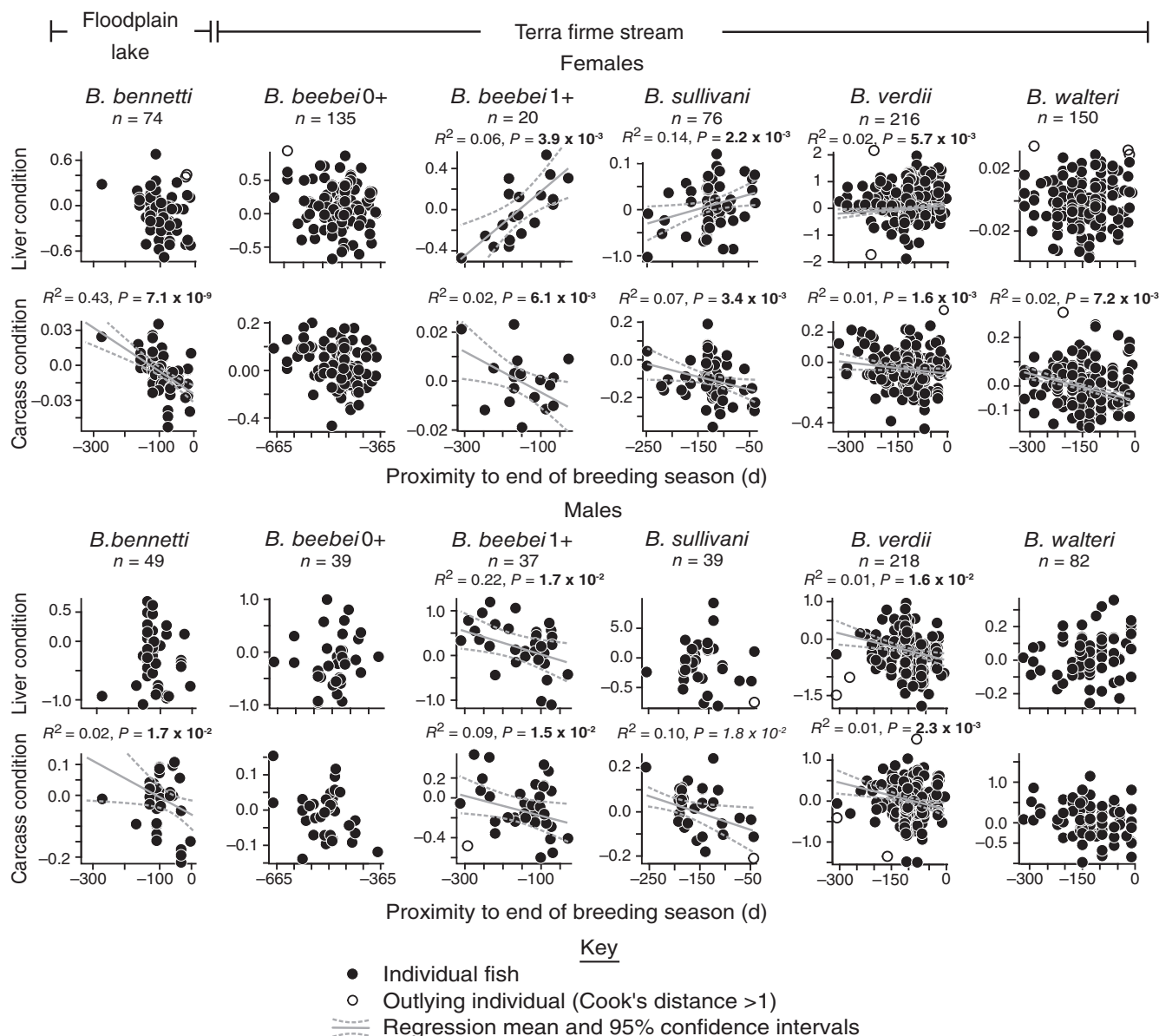


FIGURE 6 Ordinary least squares regressions of liver condition and carcass condition as a function of proximity to end of the final breeding period for spawning-capable females and males of five species of *Brachyhypopomus*. See Figure 5 legend for details

was observed in 14 out of 20 regressions, while the remaining six regressions were nonsignificant. The probability of observing this combination of slopes in the expected direction, by random chance, can be modeled by binomial theory. The binomial probability coefficient $P_{(x,n,\pi)}$ is calculated by $n!/(x!(n-x)!\pi^x(1-\pi)^{n-x})$, where x is the number of favorable outcomes, n is the number of “trials,” and π is the probability of a favorable outcome for a trial (here we assumed the probability, π , that a random distribution of values has a positive slope is 0.5). For somatic condition we calculated $P_{(10,10,0.5)}$ (henceforth P_{SC}) for the observed 10 out of 10 negative slopes as 9.766×10^{-4} ($\text{dbinom}(10, \text{size} = 10, \text{prob} = 0.5)$ in R). For gonad condition and EOD condition, a conservative

estimate of $P_{GC,EOD}$ was calculated as 0.0577, by the probability of observing 14 or more positive slopes ($\sum_{n=14}^{20} n!/(x!(n-x)!\pi^x(1-\pi)^{n-x})$ ($1 - \text{pbinom}(13, \text{size} = 20, \text{prob} = 0.5)$ in R). The value for $P_{GC,EOD}$ represents a range of possibilities, where from zero to all of the six nonsignificant slopes (which in theory could be positive or negative) are in the expected positive direction. Finally, the overall probability P that the combination of expected slope signs reported in Figure 5 could have arisen by chance alone, was calculated as P_{SC} multiplied by $P_{GC,EOD}$ ($9.766 \times 10^{-4} \times 0.0577$) = 5.631×10^{-5} .

We included outliers (individuals with Cook's distance > 1) in the analyses in Figures 5 and 6 (see white dots). Following removal of these outliers, we recovered

TABLE 1 Results of linear mixed models for Model 1: condition factor $\sim$ proximity to end of breeding season $\times$ sex + (1|species)

							Slope				
Condition factor, sex	Proximity <i>F</i>	Sex <i>F</i>	<i>R</i> ² _m	<i>R</i> ² _c	Interaction <i>P</i>	Intercept	Value	Sign	95% CIs		<i>P</i>
									Lower	Upper	
Gonad	68.0	0.5	0.0726	0.0729	0.45						
F						0.449	3.45×10^{-3}	+	4.40×10^{-3}	2.50×10^{-3}	7.81×10^{-13}
M						0.365	2.90×10^{-3}	+	2.30×10^{-3}	3.50×10^{-3}	1.25×10^{-6}
EOD	21.6	0.07	0.0224	0.0225	0.72						
F						0.069	5.27×10^{-4}	+	8.30×10^{-4}	2.24×10^{-4}	3.82×10^{-6}
M						0.078	6.15×10^{-4}	+	9.89×10^{-4}	2.41×10^{-4}	1.33×10^{-3}
Somatic	24.3	0.02	0.0252	0.0253	0.77						
F						−0.054	$−4.14 \times 10^{-4}$	−	$−1.94 \times 10^{-4}$	$−6.33 \times 10^{-4}$	2.33×10^{-4}
M						−0.058	$−4.33 \times 10^{-4}$	−	$−2.63 \times 10^{-4}$	$−6.04 \times 10^{-4}$	8.46×10^{-4}
Liver	6.8	0.08	0.0078	0.0078	0.78						
F						0.081	6.20×10^{-4}	+	1.15×10^{-3}	8.95×10^{-5}	9.31×10^{-3}
M						0.067	5.01×10^{-4}	~	1.16×10^{-3}	$−1.54 \times 10^{-4}$	1.34×10^{-1}
Carcass	23.4	0.04	0.0242	0.0243	0.67						
F						−0.052	$−3.91 \times 10^{-4}$	−	$−1.73 \times 10^{-4}$	$−6.08 \times 10^{-4}$	4.67×10^{-4}
M						−0.057	$−4.66 \times 10^{-4}$	−	$−1.96 \times 10^{-4}$	$−7.35 \times 10^{-4}$	7.45×10^{-4}

Notes: Interaction/slope *P* values set in boldface/lightface text indicate $P < 0.05/P > 0.05$, respectively, following sequential Bonferroni correction ($k = 5$; for the five condition factors of a given sex). Proximity = proximity to end of breeding season. R^2_m and R^2_c are the marginal and conditional coefficients of correlation, respectively. Degrees of freedom are 1, 957 for all condition factors, with denominator rounded to nearest integer. Sample sizes: females (F) = 536, males (M) = 425. Abbreviation: EOD, electric organ discharge.

the same supported instances of terminal investment as in Figures 5 and 6, with the exception of a loss of a negative slope for somatic condition (and consequent loss of the terminal investment pattern) in female *B. walteri* (Appendix S1: Table S1). We stress that all outliers were first checked thoroughly for any measurement errors. To explore the sensitivity of our results to our method for calculating gonad condition and somatic condition (and also liver condition and carcass condition), we calculated a second measure of these condition factors in which we calculated the residuals from SMA regression of ln mass (i.e., GM, SM, LM, CM) as a function of ln(BM) (instead of as a function of ln(LEA)). OLS regressions using this alternative set of condition factors (Appendix S1: Table S2) showed the same instances of support for the TIH as reported in Figures 5 and 6.

OLS regressions: the role of the liver in reproductive allocation

Like somatic condition, carcass condition decreased as a function of proximity to the end of the final breeding season in all cases in which a statistically clear correlation was observed, in both females and males (Figure 6). Liver

condition exhibited a more sporadic and apparently sex-specific pattern. In females, liver condition increased as a function of proximity to the end of the final breeding season in three of the five species (but decreased in none). In males, liver condition *decreased* as a function of proximity to the end of the final breeding season in two of the four species (but increased in none). There were no temporal changes in carcass or liver condition during the breeding season of the *B. beebei* 0+ year group.

Linear mixed models

Model 1: condition factor $\sim$ proximity to end of breeding season $\times$ sex + (1|species)

We used Model 1 to assess whether the response of a given condition factor to proximity to the end of the breeding season was congruent across all species. To do so, species identity was treated as a categorical random effect variable, following the rationale of van Rijn et al. (2016) and Kuehne et al. (2020). Proximity to the end of a given species' breeding season was designated as a continuously variable fixed effect, and sex as a categorical fixed effect (allowing for interaction between the fixed effects). For both females and males, Model 1 revealed a

TABLE 2 Results of linear mixed models for Model 2: condition factor $\sim$ proximity to end of breeding season $\times$ species + (1|sex)

Condition factor, species	Proximity F	Species F	R^2_m	R^2_c	Interaction P	Slope				
						Intercept	Value	Sign	95% CIs Lower Upper	P
Gonad	4.0	1.9	0.0815	0.0822	1.23×10^{-1}					
<i>B. bennetti</i>						3.98×10^{-1}	3.53×10^{-3}	+	5.99×10^{-3} 1.06×10^{-3}	5.12×10^{-3}
<i>B. beebei</i>						1.09	6.75×10^{-3}	+	9.45×10^{-3} 4.05×10^{-3}	1.14×10^{-6}
<i>B. sullivani</i>						4.76×10^{-1}	3.94×10^{-3}	+	7.41×10^{-3} 4.66×10^{-4}	2.64×10^{-2}
<i>B. verdii</i>						4.00×10^{-1}	3.29×10^{-3}	+	4.43×10^{-3} 2.14×10^{-3}	2.53×10^{-8}
<i>B. walteri</i>						3.89×10^{-1}	2.67×10^{-3}	+	4.14×10^{-3} 1.21×10^{-3}	4.49×10^{-5}
EOD	28.8	3.2	0.0395	0.0398	3.00×10^{-3}					
<i>B. bennetti</i>						1.39×10^{-1}	1.23×10^{-3}	+	2.02×10^{-3} 4.49×10^{-4}	2.13×10^{-3}
<i>B. beebei</i>						2.74×10^{-1}	1.70×10^{-3}	+	2.56×10^{-3} 8.39×10^{-4}	1.16×10^{-4}
<i>B. sullivani</i>						1.24×10^{-1}	1.01×10^{-3}	+	1.58×10^{-3} 4.50×10^{-4}	7.25×10^{-2}
<i>B. verdii</i>						7.72×10^{-2}	6.35×10^{-4}	+	1.00×10^{-3} 2.70×10^{-4}	6.76×10^{-4}
<i>B. walteri</i>						8.49×10^{-3}	5.84×10^{-4}	~	5.24×10^{-4} -4.07×10^{-4}	0.78
Somatic	20.7	2.0	0.0396	0.0399	1.07×10^{-2}					
<i>B. bennetti</i>						-1.16×10^{-1}	-1.03×10^{-3}	-	-8.81×10^{-4} -1.17×10^{-3}	4.31×10^{-4}
<i>B. beebei</i>						-4.22×10^{-2}	-2.62×10^{-4}	-	-1.02×10^{-4} -4.21×10^{-4}	2.47×10^{-3}
<i>B. sullivani</i>						-1.03×10^{-1}	-8.67×10^{-4}	-	-4.57×10^{-4} -1.28×10^{-3}	3.45×10^{-2}
<i>B. verdii</i>						-7.65×10^{-2}	-6.30×10^{-4}	-	-4.94×10^{-4} -7.65×10^{-4}	3.70×10^{-6}
<i>B. walteri</i>						-1.12×10^{-2}	-7.69×10^{-5}	~	2.19×10^{-4} -3.73×10^{-4}	0.61

Notes: Genus is *Brachyhypopomus*. Interaction/slope P values in boldface/italic/lightface text indicate $P < 0.05/P > 0.05 < 0.1/P > 0.1$, respectively, following sequential Bonferroni correction ($k = 5$; for the five condition factors). Proximity = proximity to end of breeding season. R^2_m and R^2_c are the marginal and conditional coefficients of correlation, respectively. Degrees of freedom are 4, 951 for the fixed effects, with denominator rounded to nearest integer. Sample sizes: females = 536, males = 425.

TABLE 3 Results of linear mixed model for Model 3: condition factor $\sim$ proximity to end of breeding season + (1|species/sex), considering the entire breeding season

Condition factor	F	R^2_{m}	R^2_{c}	Intercept	Slope				
					Value	Sign	95% CIs		P
							Lower	Upper	
Gonad	73.6	0.0712	0.0713	4.14×10^{-1}	3.22×10^{-3}	+	3.96×10^{-3}	2.48×10^{-3}	$<2.0 \times 10^{-16}$
EOD	21.9	0.0223	0.0223	7.23×10^{-2}	5.61×10^{-4}	+	7.96×10^{-4}	3.26×10^{-4}	3.29×10^{-6}
Somatic	24.7	0.0252	0.0267	-5.54×10^{-2}	-4.33×10^{-4}	−	-2.62×10^{-4}	-6.03×10^{-4}	7.59×10^{-7}

Notes: Slope P values in boldface type indicate $P < 0.05$ following sequential Bonferroni correction ($k = 3$; for the three condition factors). R^2_m and R^2_c are the marginal and conditional coefficients of correlation, respectively. Degrees of freedom are 1, 959 for the fixed effect (proximity), with denominator rounded to nearest integer. Sample sizes: females = 536, males = 425.

species-wide pattern consistent with the TIH, in which both gonad condition and EOD condition increased as a function of proximity to the end of a species' final breeding season, while somatic condition simultaneously decreased (Table 1). For these responses we observed no difference between females and males (see interaction terms in Table 1). Mirroring the OLS regressions reported

in Figure 6, Model 1 also revealed a species-wide decrease in both female and male carcass condition, as well as a species-wide increase in female liver condition, but little consistent change in male liver condition. The results of Model 1 were unaffected by the removal of outliers (Appendix S1: Table S3) and we therefore also included outliers in all other LMMs. For all condition factors the

TABLE 4 Results of linear mixed model for Model 3: condition factor $\sim$ proximity to end of breeding season + (1|species/sex) (a) with the first two months of the breeding season excluded and (b) with the last two months of the breeding season excluded

Condition factor	F	R^2_{m}	R^2_{c}	Intercept	Slope					P
					Value	Sign	95% CIs			
							Lower	Upper		
(a) First two months excluded										
Gonad	40.2	0.0215	0.0215	6.06×10^{-1}	4.39×10^{-3}	+	5.31×10^{-3}	3.47×10^{-3}	$<2.2 \times 10^{-16}$	
EOD	11.2	0.0232	0.0234	8.62×10^{-2}	6.90E-04	+	9.81×10^{-4}	3.99×10^{-4}	3.79×10^{-6}	
Somatic	23.0	0.0267	0.0267	-7.53×10^{-2}	-6.40E-04	-	-3.78×10^{-4}	-9.01×10^{-4}	1.89×10^{-6}	
(b) Last two months excluded										
Gonad	87.1	0.0978	0.1027	3.41×10^{-1}	2.49×10^{-3}	+	3.62×10^{-3}	1.35×10^{-3}	$<2.2 \times 10^{-16}$	
EOD	5.7	0.0071	0.0115	3.54×10^{-2}	3.59E-04	+	6.54×10^{-4}	6.45×10^{-5}	1.72×10^{-3}	
Somatic	14.9	0.0177	0.0178	-4.78×10^{-2}	-3.97E-04	-	-1.96×10^{-4}	-5.98×10^{-4}	1.29×10^{-4}	

Notes: Slope *P* values in boldface type indicate $P < 0.05$ following sequential Bonferroni correction ($k = 6$; for the two sets of three condition factors). R^2_m and R^2_c are the marginal and conditional coefficients of correlation, respectively. Degrees of freedom are (a) 1, 807 and (b) 1, 861 for the fixed effect (proximity), with denominator rounded to nearest integer. Sample sizes: (a) females = 457, males = 353; (b) females = 478, males = 386.

TABLE 5 Results of linear mixed model for Model 3: condition factor $\sim$ proximity to end of breeding season + (1|species/sex) (a) including only non-actively spawning (gonad stage 3) individuals and (b) including only actively spawning (gonad stage 4) individuals

						Slope				
Condition factor	F	df	R^2_{m}	R^2_{c}	Intercept	Value	Sign	95% CIs		P
								Lower	Upper	
(a) Non-actively spawning										
Gonad	51.0	1, 497	0.0933	0.0935	3.40×10^{-1}	3.18×10^{-3}	+	4.04×10^{-3}	2.31×10^{-3}	2.94×10^{-12}
EOD	14.4	1, 406	0.0286	0.0317	6.14×10^{-2}	5.36E-04	+	8.13E-04	2.60×10^{-4}	1.67×10^{-4}
Somatic	15.5	1, 433	0.0306	0.0333	-5.53×10^{-2}	-4.19E-04	-	-2.10E-04	-6.28×10^{-4}	1.33×10^{-4}
(b) Actively spawning										
Gonad	10.7	1, 453	0.0227	0.0311	4.21×10^{-1}	2.52×10^{-3}	+	4.03×10^{-3}	1.01×10^{-3}	1.17×10^{-3}
EOD	5.5	1, 460	0.0118	0.0119	8.49×10^{-2}	5.92E-04	+	1.09×10^{-3}	9.84×10^{-5}	1.92×10^{-2}
Somatic	8.6	1, 448	0.0184	0.0222	-6.31×10^{-2}	-5.11E-04	-	-1.70E-04	-8.52×10^{-4}	3.49×10^{-3}

Notes: Slope *P* values boldface type indicate $P < 0.05$ following sequential Bonferroni correction ($k = 6$; for the two sets of three condition factors). R^2_m and R^2_c are the marginal and conditional coefficients of correlation, respectively. Degrees of freedom are given for the fixed effect (proximity), with denominator rounded to nearest integer. Sample sizes: (a) females = 266, males = 236; (b) females = 270, males = 189.

conditional R^2 value was only slightly higher than the marginal R^2 value, indicating that little variance in the relationship between a given condition factor and proximity to the end of the breeding season is associated with variation among species. We presume that the small fraction of variance attributable to species variation resulted from the homogeneity of the slopes for condition factor $\sim$ proximity to end of breeding season among species (see Table 2), from the similarity of the breeding periods among species, and because the residuals values calculated for a given condition factor varied across a similar range in all species.

Model 2: condition factor $\sim$ proximity to end of breeding season $\times$ species + (1|sex)

We used Model 2 as an alternative to Model 1 for assessing whether the response of a given condition factor to proximity to the end of the breeding season was congruent across all species, and both sexes. Species identity was treated as a categorical *fixed* effect (with an interaction with proximity to the end of breeding season), and sex as a categorical *random* effect; here, and in subsequent models, we consider only gonad condition, EOD condition and somatic condition. Mirroring the results

summarized in Figure 5, Model 2 reports positive slopes for gonad condition and EOD condition in all species (except *B. walteri* for EOD condition), and negative slopes for somatic condition in all species except *B. walteri* (Table 2). The binomial probability of obtaining the combination of slope directions (in the direction expected by the TIH) reported in Table 2 was calculated as 3.05×10^{-5} . We observed only slightly higher conditional R^2 values than marginal R^2 values in Model 2, indicating that only a small component of the variance in Model 2 is explained by variance between the sexes. The results summarized in Table 2 were also repeated in simple linear models of the form condition factor $\sim$ proximity to end of breeding season $\times$ species, performed separately for females and males (Appendix S1: Table S4).

Model 3: condition factor $\sim$ proximity to end of breeding season + (1|species/sex)

To simultaneously explore whether the responses of a given condition factor to proximity to the end of the breeding season were generalized across both sexes, as well as across all five species, we modified Model 1 by assigning sex (nested in species) as an additional random effect. For simplicity, here we considered only gonad condition, EOD condition, and somatic condition. Model 3 retained a clear pattern of terminal investment, with gonad condition and EOD condition again increasing as a function of proximity to the end of the breeding season and somatic condition decreasing (Table 3).

The analyses reported in Table 3 were based on samples for the entire breeding season and combined all spawning-capable individuals. The same patterns reported in Table 3 were repeated in analyses in which we removed individuals captured during the first two months of the final breeding season, or the last two months of each species' final breeding season (Table 4). Population-wide patterns of terminal investment were evidently not highly leveraged by the diversion of somatic resources to reproduction during the late and/or early parts of the breeding season.

The patterns reported in Table 3 were also repeated in analyses restricted to spawning-capable individuals in the non-actively spawning stage (gonad stage 3), or to individuals in the actively spawning stage (gonad stage 4; Table 5). This finding rules out the possibility that changes along the course of each species' breeding season in the ratio of actively spawning to non-actively spawning fish could lead to temporal modulations in reproductive allocation that mimic terminal investment.

Model 4: condition factor $\sim$ proximity to end of breeding season $\times$ body size (or SMI) + (1|species/sex)

We used Model 4 to explore the effect of body size (LEA) and overall body condition (SMI) on patterns of resource allocation. Each species/age/year-group combination was first divided into two body size categories (small and

TABLE 6 Results of linear mixed models for Model 4: condition factor $\sim$ proximity to end of breeding season $\times$ body size + (1|species/sex)

Condition factor, size	Proximity F	df	Sex F	df	R^2_m	R^2_c	Int. P	Slope					
								Intercept	Value	Sign	95% CIs		
											Lower	Upper	P
Gonad	71.9	1,957	0.05	1,956	0.1511	0.1758	0.94						
L								6.99×10^{-1}	3.26×10^{-3}	+	4.38×10^{-3}	2.14×10^{-3}	$<2.2 \times 10^{-16}$
S								2.72×10^{-1}	3.21×10^{-3}	+	4.19×10^{-3}	2.22×10^{-3}	$<2.2 \times 10^{-16}$
EOD	22.3	1,957	0.05	1,957	0.0230	0.0231	0.89						
L								7.01×10^{-2}	5.95×10^{-4}	+	9.59×10^{-4}	2.31×10^{-4}	2.73×10^{-6}
S								7.85×10^{-2}	5.61×10^{-4}	+	8.74×10^{-4}	2.48×10^{-4}	2.73×10^{-6}
Somatic	93.2	1,946	0.36	1,950	0.0587	0.0749	0.56						
L								-6.65×10^{-2}	-3.34×10^{-4}	-	-1.04×10^{-4}	-5.63×10^{-4}	1.86×10^{-8}
S								-4.72×10^{-2}	-6.72×10^{-4}	-	-4.11×10^{-4}	-9.33×10^{-4}	1.86×10^{-8}

Notes: Interaction (Int.)/slope P values in boldface/lightface text indicate $P < 0.05/P > 0.05$, following sequential Bonferroni correction ($k = 6$; for the two sets of three condition factors). Proximity = proximity to end of breeding season. R^2_m and R^2_c are the marginal and conditional coefficients of correlation, respectively. Degrees of freedom are given for the fixed effects, with denominator rounded to nearest integer. Sample sizes: large (L) body size (n females = 293, n males = 201); small (S) body size (n females = 243, n males = 224).

TABLE 7 Results of linear mixed models for Model 4: condition factor $\sim$ proximity to end of breeding season $\times$ scaled mass index (SMI) + (1|species/sex)

Condition factor, SMI								Slope						
	Proximity		Sex		Interaction			Intercept	Value	Sign	95% CIs			
	F	df	F	df	R ² _m	R ² _c	P				Lower	Upper	P	
Gonad	76.2	1, 956	0.8	1, 955	0.3224	0.3234	0.36							
High								7.60×10^{-1}	2.64×10^{-3}	+	3.78×10^{-3}	1.49×10^{-3}	1.10×10^{-10}	
Low								7.25×10^{-1}	1.99×10^{-3}	+	2.76×10^{-3}	1.22×10^{-3}	1.10×10^{-10}	
EOD	15.1	1, 957	0.2	1, 957	0.0240	0.0240	0.65							
High								6.95×10^{-2}	4.52×10^{-4}	+	8.81×10^{-4}	2.36×10^{-5}	1.05×10^{-4}	
Low								6.63×10^{-2}	5.73×10^{-4}	+	8.60×10^{-4}	2.86×10^{-4}	1.05×10^{-4}	
Somatic	29.2	1, 957	2.0	1, 957	0.0309	0.0310	0.14							
High								-5.61×10^{-2}	-3.75×10^{-4}	-	-1.67×10^{-4}	-5.83×10^{-4}	8.24×10^{-8}	
Low								-6.99×10^{-2}	-6.54×10^{-4}	-	-3.44×10^{-4}	-9.64×10^{-4}	8.24×10^{-8}	

Notes: Interaction (Int.)/Slope P values boldface/lightface text indicate $P < 0.05/P > 0.05$, following sequential Bonferroni correction ($k = 6$; for the two sets of three condition factors). Proximity = proximity to end of breeding season. R^2_m and R^2_c are the marginal and conditional coefficients of correlation, respectively. Degrees of freedom are shown for the fixed effects, with denominator rounded to nearest integer. Sample sizes: low SMI, i.e., low condition (n females = 262, n males = 211); high SMI (n females = 274, n males = 214).

large, divided at the mean LEA) or two body condition categories (low and high, divided at the mean SMI). Body size and SMI were then assigned as fixed effect categorical predictor variables (allowing interaction between the fixed effects).

We observed no difference between small and large individuals in the slope of the positive covariance between gonad condition (or EOD condition) and proximity to the end of the breeding season, nor did we find a slope difference between small and large individuals in the negative covariance of somatic condition with proximity to the end of the breeding season (Table 6). Likewise, for the equivalent analyses, we observed no difference between individuals with low and high SMI (Table 7). The patterns we report in Tables 6 and 7 were also found in LMMs in which males and females were considered separately (Appendix S1: Tables S5 and S6).

DISCUSSION

We documented support for the terminal investment hypothesis by an increase in two metrics of reproductive effort (size-adjusted gonad mass and size-adjusted EOD signal amplitude), as well as a concomitant decrease in somatic condition (size-adjusted mass of the body minus the gonads) indicative of a cost to future survival and reproduction, all in response to proximity to the end of a species' final breeding season. This shift in reproductive allocation occurs synchronously in all the *Brachyhyppomus* species from our upper Amazon study

sites, including in the females of all five species and in the males of four species.

In both floodplain and terra firme habitats, the end of the *Brachyhyppomus* breeding season is coincident with the onset of a dry-season period of greatly elevated predation and reduced habitat availability (Henderson et al., 1998; Lowe-McConnell, 1987; Röpke et al., 2019; Waddell & Crampton, 2020). Therefore, given that the end of the final breeding season of *Brachyhyppomus* is followed by complete post-reproductive mortality (Waddell et al., 2019; Waddell & Crampton, 2020), and thus corresponds to the point at which their RRV reaches zero, our results unequivocally demonstrate terminal investment in response to an outright diminution of RRV. Moreover, our recovery of a pattern consistent with terminal investment in the final (1+) but not first (0+) year group of the biseasonal species *B. beebei* demonstrates that the diversion of resources from the somatic condition to reproductive effort is a specific response to a critical reduction of RRV, and not a fixed response to seasonal environmental variation, per se. Finally, we stress that all elements of body mass were accounted for in our measurements of gonad condition (gonad mass) and somatic condition (total body mass minus gonad mass). Therefore, our observations of increasing gonad condition concomitant with decreasing somatic condition as a function of proximity to the end of a species' final breeding season are precisely indicative of the pattern expected by the TIH, that is, a generalized, population-wide diversion of resources from the soma to the gonads as RRVs decline toward zero.

We are confident that the design of this study has permitted robust diagnosis of the terminal investment response in *Brachyhypopomus*. First, we quantified metrics of both reproductive effort and somatic depletion only from reproductively mature individuals, for which we obtained large sample sizes for each species, sex, and year group. Second, addressing concerns by Roff (2002) for body mass metrics and also Gavassa et al. (2013) for EOD amplitude, we made appropriate corrections for body size in the calculation of all of our condition metrics. Third, addressing concerns by Clutton-Brock (1984), Velando et al. (2006), and Javoš (2013), we diagnosed cases of terminal investment only where we identified both a statistically clear elevation in metrics of reproductive effort, and a concomitant reduction in somatic condition indicative of the requisite survival cost levied against future reproductive events. This third consideration led us to conclude that terminal investment may occur more sporadically in male than in female *Brachyhypopomus* (and may be entirely absent in *B. walteri*). Nonetheless, many other studies of reproductive life history have taken a less conservative approach by interpreting elevations of reproductive effort in later life stages as de facto support for the TIH (even if no measurable diminution of somatic condition is reported), based on the assumption that increased reproduction is intrinsically costly (e.g., Lafaille et al., 2010).

By verifying that there is a survival cost to an elevation in reproductive effort, we are specifically able to rule out some processes that can mimic terminal investment (Duffield et al., 2018). These include a decrease in the late-season survival of low-performing reproducers (Mauck et al., 2004; Service, 2000), or the tendency for older individuals to be better able to intensify reproductive effort than younger individuals under deteriorating environmental conditions (Duffield et al., 2018); neither of these processes alone would predict a population-wide diminution of somatic condition concomitant with an elevation of reproductive effort.

Our study design nonetheless does have one important limitation in that, although we were able to differentiate between individuals belonging to the 0+ and 1+ year groups of *B. beebei* (via size distributions and sagittal otolith annuli), we were unable to determine the exact chronological age (i.e., days since hatching) of the individuals of any *Brachyhypopomus* species. While we found annual growth rings in individual otoliths, we did not find clear daily growth rings. Therefore, although we were able to observe a distinct first vs. second year difference in the reproductive life history of *B. beebei*, we were unable to disentangle fine-scale age from other predictors of RRV (such as extrinsic environmental variation, or intrinsic factors such as condition) as potential triggers of

the shift in reproductive investment in *Brachyhypopomus* (see *Cues for terminal investment*).

Another limitation of our study is that we were not able to document all possible measures of reproductive effort and survival cost in the *Brachyhypopomus* system. Measures of reproductive effort that we did not quantify include motor behavior associated with courtship and intrasexual contests, as well as potentially energetically costly EOD pulse rate modulations associated with sexual interactions (Hagedorn & Heiligenberg, 1985). Measures of survival cost that we did not measure include immune health, which is typically negatively correlated with reproductive effort in wild animal populations (e.g., Gustaffson et al., 1994), especially in males (Klein & Flanagan, 2016). Additionally, although we treated EOD amplitude here as a metric of reproductive effort, conspicuous EODs can potentially incur a simultaneous survival cost, which we did not measure, by attracting eavesdropping electroreceptive predators (Crampton et al., 2013; Stoddard, 2002).

Terminal investment in uniseasonal iteroparous species

Based on our criteria for diagnosing terminal investment, we documented the phenomenon in all uniseasonal iteroparous (annual) species of *Brachyhypopomus*. Nonetheless, although support for terminal investment was robust in both sexes of *B. bennetti*, *B. sullivani*, and *B. verdii*, we failed to report the phenomenon in males of *B. walteri*. Support for terminal investment was also weak in female *B. walteri* (with outliers removed from the OLS regressions in Figure 5, the diagnostic pattern was lost). The general reproductive life history of *B. walteri* resembles those of the other uniseasonal iteroparous congeners and we are therefore unable to explain why the terminal investment response is expressed weakly in this species.

The patterns we observed in uniseasonal iteroparous *Brachyhypopomus* species resemble those reported by Poizat et al. (1999) for females of an annual population of three-spined sticklebacks from a strongly seasonal wetland. Like us, Poizat et al. (1999) diagnosed terminal investment as a population-wide pattern of positive covariance between size-adjusted ovary mass (our GC) and proximity to the end of the single breeding season, as well as a concomitant negative covariance between size-adjusted somatic mass (the authors used an equivalent to our CC) and proximity to the end of the breeding season. Patterns of resource reallocation consistent with terminal investment have also been reported from studies involving experimental manipulations of condition in uniseasonal iteroparous insect taxa, including Coleoptera,

Orthoptera, and Diptera (Duffield et al., 2017), in the *Brachyhypopomus* species *B. gauderio* (Gavassa & Stoddard, 2012), and possibly also in a nearly unisexual iteroparous rodent, the white-footed mouse (Derting & Virk, 2005). Nonetheless, the effects of seasonality were not explored in these studies.

Terminal investment in multiseasonal iteroparous species

As predicted by life history theory, we reported a pattern of resource reallocation consistent with terminal investment in the second (1+) and final breeding season of the multiseasonal iteroparous *B. beebei*, but not in the first (0+) breeding season. In the first breeding season, we instead reported a contrary pattern of reproductive restraint manifested by relatively constant reproductive effort, and no somatic depletion. Our results resemble those of Billman and Belk (2014), who reported increasing size-adjusted gonad mass and decreasing size-adjusted somatic mass with proximity to the end of a second (but not first) breeding season in a multiseasonal iteroparous mosquitofish population with a 2-year life span. Terminal investment in the final breeding season (but not earlier breeding seasons) of some longer-lived iteroparous species has been reported, for example in Wandering Albatrosses (Froy et al., 2013) and Banded Wrens (Hall et al., 2009). However, terminal investment (where observed) is more commonly generalized across breeding seasons later in life, but not entirely confined to the final season, for example, in red squirrels (Descamps et al., 2007), Nile lechwe (Bercovitch et al., 2009), and moose (Ericsson et al., 2001).

The change in the allocation of reproductive effort between the 0+ and 1+ year groups of *B. beebei* is consistent with a switch from an “income breeding” strategy (sensu Jönsson, 1997), in which reproductive effort is sustained solely by feeding, to a “capital breeding” strategy, in which an increase in reproductive effort is supplemented by the depletion of somatic reserves. Moderating reproduction in the first year is expected to increase the prospects of survival to the second year, at which point individuals should benefit from increased body size. For fish in general, body size is positively correlated to fecundity (Parker, 1992). In electric fish, the EOD plays an important role in the reproductive signaling of both sexes, and because of the exponential positive covariance of EOD amplitude with body size, even small increments in the growth of larger individuals can augment breeding success. For example, in both male and female *Brachyhypopomus* spp., larger body size and higher EOD amplitude are associated with increased

success in intrasexual conflicts (Gavassa et al., 2012; Hagedorn & Zelick, 1989; Stoddard, 2006). Large male *Brachyhypopomus gauderio* also appear to be more attractive to sexually mature females than small males (Curtis & Stoddard, 2003).

The time scale of terminal investment

Discussions of the TIH tend to make no special discrimination between two scenarios: (1) elevated reproductive investment in later year groups relative to earlier year groups (in multiseasonal iteroparous species) and (2) elevated reproductive investment in later reproductive events within a year group relative to earlier events in the same year group (in unisexual iteroparous species). The difference between scenarios 1 and 2 is essentially one of time scale only, because both involve a diversion of resources from the soma to reproduction in response to a critical lowering of RRV. Nonetheless, we argue that a distinction between these two scenarios becomes important in multiseasonal animals with short lifespans, particularly in biseasonal species, where the duration of an individual's final year group represents a larger proportion of its total lifespan. For example, in *B. beebei*, both scenarios occur simultaneously; we observed not only the theoretically predicted elevation of reproductive effort in the second year relative to the first year, but we also observed an elevation of reproductive effort in late season reproductive events in the second season, relative to earlier events. Biseasonal iteroparous animals like *Brachyhypopomus beebei* and some mosquitofish populations (Billman & Belk, 2014) therefore show elements of the pattern of terminal investment seen in long-lived animals such as many birds and mammals, but also elements of the pattern of terminal investment in insects and other annual animals (Duffield et al., 2017). We conclude that terminal investment occurs over a continuum of time scales, from annual species through short-lived multiseasonal species, to long-lived multiseasonal species, but in all cases via a phenomenologically equivalent response to diminishing RRV.

The disposable soma theory

Our observation of a clear disparity in patterns of reproductive allocation between closely related species with unisexual iteroparous life histories vs. multiseasonal iteroparous life histories provides compelling support for Kirkwood's (1987) disposable soma theory. This theory postulates that death by reproductive exhaustion can be delayed by moderating reproductive effort in a given breeding season, thereby allowing individuals to survive

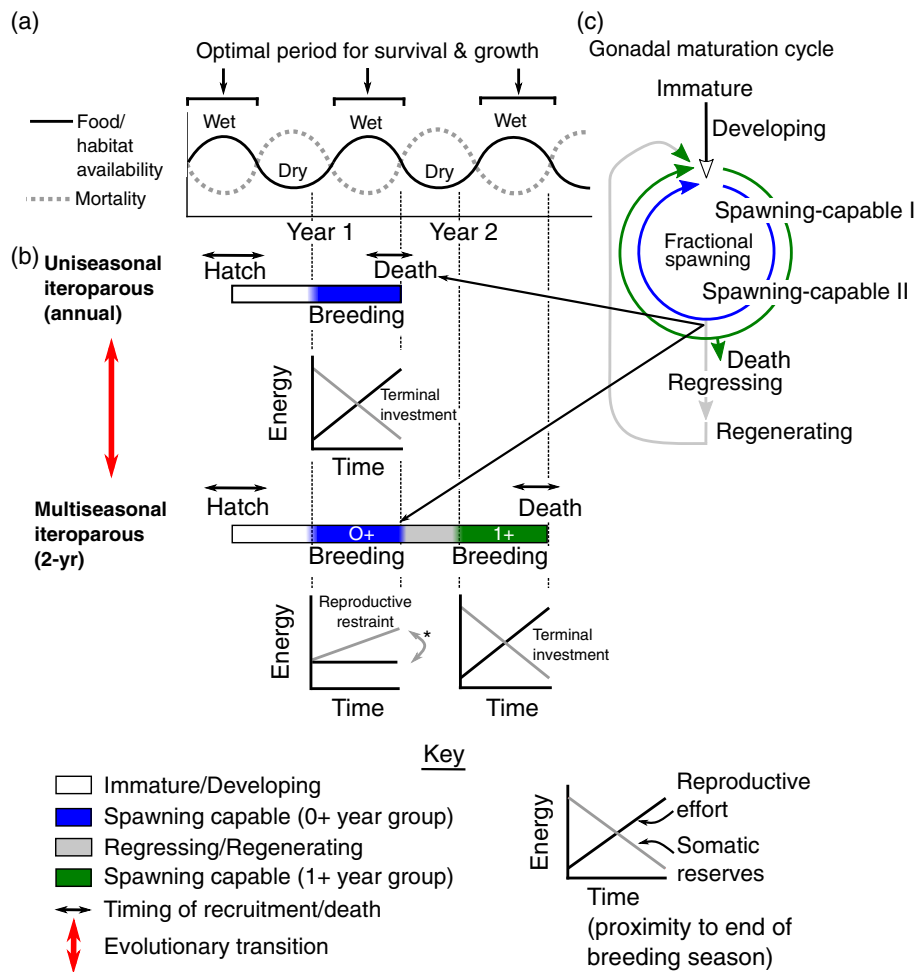


FIGURE 7 A conceptual model for terminal investment in short-lived aquatic animals from highly seasonal habitats. Seasonal variation in food/habitat availability and predator/disease-induced mortality (a) enforces reproduction during a single prolonged annual breeding season, timed to a wet-season period of optimal recruitment success and growth. In an evolutionary transition from uniseasonal iteroparity to 2-year multiseasonal iteroparity (b), species substitute a pattern of terminal investment (in which energy is diverted from somatic growth and maintenance to reproduction in their first, 0+, year group) with a pattern of reproductive restraint, in which breeding effort is moderated in favor of somatic growth. The double arrow marked with an asterisk denotes that, in 0+ individuals, somatic condition is predicted to either covary positively with time (proximity to the end of the breeding season), or to show no covariance (but not to covary negatively with time). For oviparous fish, a transition from uniseasonal to multiseasonal iteroparity also requires a modification of the typical gonadal maturation cycle (c) via the origin of a sequence of gonadal regression in 0+ individuals followed by gonadal regeneration at the onset of the 1+ breeding season (represented here by the arrow fading from blue to green, via gray). Gonadal maturation stages follow Brown-Peterson et al. (2011). Individuals in both the 0+ and 1+ year groups alternate between spawning-capable I status (stage 3: non-actively spawning) and spawning-capable II status (stage 4: actively spawning) via successive fractional spawning events, throughout the breeding season; this alternation of gonadal status is represented here by the green and blue circular arrows. Figure modified from Waddell et al. (2019)

and breed in a later season. In turn, increased reproductive success associated with larger body size in a later season can compensate for the costs of diminished early-life reproductive effort, as well as the costs of elevated mortality during the typically inhospitable period between successive breeding seasons.

As summarized in the conceptual schematic in Figure 7, the disposable soma theory provides a framework for understanding the processes influencing

evolutionary transitions between uniseasonal iteroparity (sometimes regarded as a form of semelparity; Kirkendall & Stenseth, 1985) and 2-year multiseasonal iteroparity (“full” iteroparity). Specifically, we propose that 2-year multiseasonal iteroparity can evolve from an annual life history via two innovations: first, the replacement of terminal investment in the first breeding season by a pattern of reproductive restraint (Figure 7b); second, the origin of a sequence of gonadal regression and regeneration during

the non-breeding period between the first and second breeding seasons (Figure 7c). By extension of the same argument, a biseasonal species with terminal investment followed by post-reproductive mortality in its second year could extend its lifespan to a third year by substituting a pattern of elevated reproductive investment in the second breeding season with reproductive restraint, and so on.

We assume that both uniseasonal and multiseasonal iteroparous life history strategies in *Brachyhypopomus* can constitute evolutionarily stable strategies in a variety of seasonal habitats. However, in a scenario where seasonal breeding conditions fluctuate markedly from year to year, multiseasonal iteroparity, in the long run, is expected to be more successful than uniseasonal iteroparity because it spreads the risk of failed reproductive events over more than one season (Wilbur & Rudolf, 2006; Wittenberger, 1979). For example, three-spined stickleback populations in ephemeral habitats are known to exhibit uniseasonal iteroparity, while populations in perennial habitats tend to exhibit multiseasonal iteroparity (Poizat et al., 1999; Wootton, 1984).

We are unaware of population-level variation in the lifespan of *Brachyhypopomus*, but of the 17 species of *Brachyhypopomus* from Greater Amazonia (the Amazon, Orinoco, and Guiana drainages), 15 show size distributions consistent with an annual, uniseasonal life history, which is the ancestral condition (Crampton et al., 2016b). In contrast, only two species, *B. beebei* and *B. brevirostris*, exhibit a derived 2-year multiseasonal life history (Crampton et al., 2016a, 2016b; Waddell et al., 2019). *Brachyhypopomus beebei* and *B. brevirostris* occupy both of the major habitats occupied by the genus, terra firme stream and floodplain habitats, while only three of the 10 annual species have a similar eurytopic habitat distribution (Crampton et al., 2016b). *Brachyhypopomus beebei* and *B. brevirostris* are also unusual in occupying broad latitudinal ranges across most of Greater Amazonia (Crampton et al., 2016a) and in occupying all three major biogeochemical river types of the lowland Neotropics: blackwater, whitewater, and clearwater (Bogotá-Gregory et al., 2020). These patterns corroborate the notion that a 2-year lifespan may provide more consistent reproductive and ecological success than an annual life history strategy over a broad range of seasonal hydrological regimes and habitats.

Terminal reproductive signaling

We report a terminal elevation of size-adjusted EOD amplitude, a quantitative measure of reproductive signaling cost, in the final breeding season of multiple *Brachyhypopomus* species. Hall et al. (2009) reported a

terminal increase in reproductive signaling in a wild population of Banded wrens, but to the best of our knowledge, this phenomenon has not been documented in other studies of natural animal populations (that is, in the absence of experimental manipulations). Nonetheless, an elevation of costly reproductive signaling consistent with terminal investment theory has been induced in several taxa by experimental manipulation of condition, for example in the acoustic calls of male Japanese tree frogs (via pathogenic infection) (An & Waldman, 2016), in the pheromonal signals of mealworm beetles (following simulated infection; Sadd et al., 2006), in the nuptial colors of male three-spined sticklebacks (via calorific restriction; Candolin, 1999), and also in the EOD amplitude and EOD duration of the *Brachyhypopomus* species *B. gauderio* (again via calorific restriction; Gavassa & Stoddard, 2012). Mathematical models presented by Kokko (1997) and Lindstrom et al. (2009) also provide a theoretical framework for how males may benefit from reallocating energy from somatic maintenance and growth to reproductive signaling, as their RRV declines.

We documented an elevation of the intensity of a female reproductive signal consistent with terminal investment, a novel result. In most animals, reproductive communication signals are typically only generated in males (Bradbury & Vehrencamp, 2011); although see Riebel et al. (2019) for bird song. However, in electric fish, both sexes (and also immature individuals) generate EODs constantly. This is because the EOD's primary function as an active electrolocation signal (Crampton, 2019) has allowed the EODs of female (as well as male) electric fish to serve a secondary role in reproductive signaling, including in female intrasexual contests (Gavassa et al., 2012) and possibly also as female epigamic signals (Crampton et al., 2011). In general, female sexual signaling is an overlooked topic (Riebel et al., 2019), and one that may prove to be especially amenable to experimental investigation in electric fish.

A clear pattern of elevated late-season EOD condition was observed in the females of four of the five *Brachyhypopomus* species, but in the males of only two species (Figure 5). Although the wide-scale elevation of late-season size-adjusted electric signal amplitude in females is consistent with the known sexual signaling role for female EODs, we were surprised that these responses were more prevalent in females than males. The disparity could be related to sex differences in the relative energetic costs of the reproductive EOD signal. Salazar and Stoddard (2008) estimated that electric signaling constitutes a substantive 11%–25% of the total 24-h energy budget of reproductive male *B. gauderio*, but a more modest 2%–4% in reproductive females. Because of the very high cost of the male EOD, and because male EOD amplitude

is known to be highly contingent on immediate social contexts, especially proximity to rival males (Franchina et al., 2001), we suspect that individual males, as an energy-saving measure, may elevate their nocturnal EOD amplitude on a more intermittent basis than females. Consequently, in comparison to females, we may have encountered a smaller proportion of late-season males that were expressing high EOD amplitude at the exact time that we captured and recorded them. This effect would weaken correlations of EOD condition with proximity to the end of the breeding season, and therefore explain why EOD condition is less clearly elevated in males captured later in the breeding season than in females.

The role of the liver in terminal investment

Studies of terminal investment seldom consider the metabolic pathways by which somatic resources are diverted to reproductive processes. In fishes, these pathways are complicated by the sexually dimorphic role of the liver (Chellappa et al., 1995). The fish liver is a multi-function organ, which in both sexes plays important roles in digestion, detoxification, nitrogen catabolism, and short-term lipid energy storage (Allen & Wootton, 1982; Bruslé & González i Anadon, 1996); longer term energy stores in fish are typically accumulated in visceral fat and the skin. However, in females only, the liver performs an important role in gametogenesis by serving as the primary site for synthesis of the yolk protein precursor vitellogenin, which is essential for oocyte production (Bruslé & González i Anadon, 1996).

By partitioning somatic condition into liver condition and carcass condition, our analyses provide some insights into the role of the liver during terminal investment. The male *Brachyhypopomus* liver appear to have a more variable role than the female liver: in some species it serves as a depletable energy store, but in other species (and summed across all species in the LMM in Table 1), it exhibits no change in size-adjusted mass through the breeding season. In contrast, the female liver appears to function not as a net depletable energy store but instead as a form of “accessory reproductive organ” to the gonads, providing the vitellogenin required for oocyte production and increasing in size-adjusted mass in lock-step with size-adjusted gonad mass. The primary energy reserves supporting terminal investment in females must instead be located in other somatic locations, such as in cutaneous, visceral, and electric organ-associated fat, or in muscle. In light of these considerations, we argue that the liver of female (but not male) *Brachyhypopomus* is best regarded as a metric of

reproductive effort rather than as a metric of somatic depletion, and that the positive covariance between female liver condition and proximity to the end of the breeding season (Figure 6, Table 1) therefore constitutes robust additional evidence for the terminal investment response in *Brachyhypopomus*.

Cues for terminal investment

Duffield et al. (2017) proposed a “dynamic threshold model” in which the triggers for elevated terminal investment typically involve context-specific interactions between intrinsic factors (such as age and condition) and extrinsic factors such as predation risk. We found an age-dependency to terminal investment in *Brachyhypopomus*, based on the terminal investment response observed in the second but not first year-group of *B. beebei*. However, although terminal investment is clearly synchronized to circannual breeding periodicity, we were unable to elucidate the specific cues responsible for triggering and sustaining the population-wide pattern of terminal investment observed in the final year of both unisexual iteroparous and multiseasonal iteroparous species.

Terminal investment could, in principle, be triggered by environmental cues. Seasonal environmental variation is known to influence the duration and timing of breeding periods in tropical freshwater fish (Munro, 1990) and to also affect reproductive trade-offs by altering resource availability and age-independent mortality risk (McNamara & Houston, 1996). Waddell and Crampton (2020) used generalized linear models and model selection to demonstrate that the onset of breeding in upper Amazon *Brachyhypopomus* species is correlated with rising river level (and to some extent increasing pH) in floodplain species, and with increasing rainfall in terra firme species, but is otherwise uninfluenced by physicochemical water properties. Waddell and Crampton (2020) also concluded that *Brachyhypopomus* likely respond to these environmental cues during a sensitive period in early gonadal maturation (or in the case of 1+ year group *B. beebei*, during gonadal recrudescence) and thereafter continue to breed regardless of changes in environmental conditions outside this sensitive period.

There are also no clear indications of how *Brachyhypopomus* could unambiguously “predict” the end of their final breeding season from common environmental cues. For example, long-term averages of the river level and local rainfall for the study area reach their maxima approximately two and four months, respectively, before the end of the annual breeding season (two months and one month, respectively, in the study year) and thereafter decline steadily in a manner predictive of the

forthcoming dry season (Appendix S3: Figure S1). However, these late-breeding-season declines in river level and rainfall cannot alone be responsible for the observed terminal investment response because we recovered robust patterns of terminal investment even with the last two months of the breeding seasons removed from our analyses (Table 4b). In summary, we find it unlikely that *Brachyhypopomus* can elevate reproductive allocation in response to increasing water level, rainfall, and habitat availability (i.e., changes indicative of increasingly *favorable* conditions) during the early part of the breeding season, because they continue to elevate reproductive allocation in the later part of the season when these same conditions reverse.

Brachyhypopomus could, alternatively, terminally invest in response to a lowering of condition resulting from decreasing food availability through the breeding season, for instance, due to intraspecific competition (Röpke et al., 2019), interspecific competition among the many species of fish that breed during the rainy season (Lowe-McConnell, 1979), or due to diminishing substrate availability (Espírito-Santo & Zuanon, 2016; Hess et al., 2003; Junk, 1970; Piedade et al., 2010). Lowered condition could also occur in association with increased pathogen transmission (Crampton, 2008). Gavassa and Stoddard (2012) reported substantial elevations of EOD amplitude in *Brachyhypopomus* following a relatively short period of food restriction (~10 d), suggesting that the elevations of EOD amplitude we observed could, in principle, be triggered by food deprivation, prior to any appreciable loss of body condition. However, summed across all species and both sexes, we reported no difference in the strength of the terminal investment response between individuals with high vs. low scaled mass indices (SMI), which we used as a metric of general body condition (Table 7). This finding contrasts with Poizat et al.'s (1999) study of an annual stickleback population, in which high-condition females (but not low-condition females) exhibited positive covariance between size-adjusted gonad mass and proximity to the end of the breeding season (low-condition females exhibited no covariance). Poizat et al. (1999) concluded that the potential for female sticklebacks to increase ovary mass is realized only in individuals with sufficient somatic reserves. Our results, on the contrary, indicate that the population-wide diversion of resources from the soma to reproductive effort in *Brachyhypopomus* is not contingent on general body condition.

Based on the absence of obvious environmental or intrinsic condition-related cues for the population-wide terminal investment response, we instead conjecture that *Brachyhypopomus* come into spawning-capable condition as a result of rainfall and/or river level cues (Waddell & Crampton, 2020) and thereafter exhibit a “fixed pattern” of

terminal investment throughout their final breeding season, one in which they gradually divert resources from the soma to reproduction, via successive reproductive events, as a simple function of their chronological age. Some support for such a fixed pattern of terminal investment comes from the observation of a similar terminal investment pattern in small versus large individuals within a given year group (Table 6). Because small and large individuals correspond approximately to younger and older individuals, respectively, the observation implies that terminal investment occurs throughout the entire final breeding season of a given species. A fixed terminal investment response would hypothetically have the advantage of being non-dependent upon potentially unreliable external or internal estimators of RRV, but would result in a less efficient deployment of lifetime reproductive effort in late-recruiting individuals (i.e., those hatching from late-breeding parents in the previous breeding season) than in early-recruiting individuals; all else being equal, late-recruiting individuals should have fewer reproductive opportunities before the end of the common breeding season.

Further studies of terminal investment in *Brachyhypopomus* would benefit from experiments at the individual level, in which the precise age of captive-reared individuals is monitored. For example, a fixed terminal investment response, in which reproductive effort is progressively augmented as a function of age, could be tested by tracking the EOD amplitude of captive males and females across multiple spawning events; this parameter can be measured from live fish without the need for dissection. Likewise, classical experimental manipulations of condition (e.g., via food restriction or simulated infection) may provide further means to understand the relative importance of fine-scale age, extrinsic mortality-inducing processes, and internal condition in regulating the terminal investment response. The results of this study provide a framework for future experiments.

CONCLUSIONS

The multispecies and multi-habitat approach of this study revealed a congruent, species-wide, and habitat-wide pattern of terminal investment that is synchronized by seasonal breeding in response to circannual environmental variation. Our results suggest that the timing of seasonal breeding (in response to variation in mortality risk), in combination with age (at the scale of year-groups) can influence the timing of resource allocation to current reproduction vs. to somatic condition and survival. We also demonstrated that the substitution of a pattern of terminal investment with a pattern of reproductive

moderation can facilitate survival and reproduction into a subsequent breeding season, thereby extending reproductive lifespan into a successive year.

We have not yet recovered a clear mechanism by which *Brachyhypopomus* could use intrinsic cues or extrinsic environmental cues as predictors of proximity to the end of their final breeding season. Nonetheless, the genus may prove to be a fruitful model for further studies of reproductive life history and of the mechanisms that stimulate and maintain terminal investment responses in natural animal populations. In addition to the ease by which metrics of reproductive effort can be measured in quantitative population-level field surveys, several species are amenable to aquarium-based rearing (Kirschbaum & Schugardt, 2002), providing opportunities to corroborate the patterns we report herein at the individual level. Future studies are also expected to benefit from the tools and resources developed during three decades of behavioral and endocrinological investigations of the neuroethological model species, *B. gauderio* (reviewed by Gavassa et al., 2013).

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CONFLICT OF INTEREST

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

Data (Waddell & Crampton, 2021) are available from the Dryad Digital Repository: <https://doi.org/10.5061/dryad.xgxd254hd>.

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SUPPORTING INFORMATION

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